Health Technology Assessment & Capital Allocation Framework

Nova Scotia Department of Health and Wellness

- **Classification:** Strategic Policy Document
- **Target Audience:** Director of Planning and Investment, Deputy Minister, Treasury Board
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Executive Summary & Mandate

The Nova Scotia Department of Health and Wellness (DHW) faces the dual challenge of accelerating clinical innovation while managing strict fiscal boundaries and environmental constraints. This framework operationalizes the province's *Action Plan for Health* by providing a standardized, repeatable, and data-driven methodology for evaluating health technology investments.

By combining frontline clinical metrics with rigorous lifecycle costing, this framework removes subjectivity from capital allocation. It ensures that every provincial dollar spent on medical devices, digital health platforms, and facility infrastructure maximizes patient outcomes, advances health equity, and preserves long-term system sustainability.

Core Governance & Implementation Workflow

To ensure objective evaluation, every capital request exceeding $250,000 must progress through the following four-stage lifecycle before submittal to the Treasury Board:

[Stage 1: Intake] —> [Stage 2: Technical Assessment] —> [Stage 3: MCDA Scoring] —> [Stage 4: Governance Review]

Stage 1: Strategic Intake & Alignment

- Submitting health zones (Central, Northern, Western, or Eastern) submit a formal Project Charter.
- Initial screening verifies that the technology complies with Health Canada regulations and Nova Scotia cyber-security baseline standards.

Stage 2: Multi-Disciplinary Technical Assessment

- **Clinical Review Panel:** Evaluates frontline clinical utility, workload impacts, and safety data.

- **Financial Review Panel:** Builds total cost of ownership (TCO) models and evaluates procurement options (e.g., capital purchase vs. Software-as-a-Service).

Stage 3: Multi-Criteria Decision Analysis (MCDA)

- The project is scored against the formal HTA Matrix.
- Scores are weighted and normalized to generate a definitive Strategic Priority Score (0–100).

Stage 4: Governance Review & Sign-Off

- The Director of Planning and Investment reviews the prioritized portfolio.
- Recommendations are forwarded to executive leadership for funding authorization.

Comprehensive Evaluation Rubric

Pillar 1: Clinical Quality and System Capacity (Weight: 30%)

Sub-Metric	Evaluation Criteria	Score 1-4 (Low Impact)	Score 5-7 (Moderate Impact)	Score 8-10 (High Impact)
Clinical Efficacy & Safety *(Weight: 10%)*	Reductions in adverse events, diagnostic errors, or re-admission rates.	Marginal change over current baseline; weak clinical evidence.	Incremental safety gains; supported by mid-tier peer-reviewed data.	Paradigm shift in care; validated by high-quality RCTs or systemic reviews.
Frontline Workflow Impact *(Weight: 10%)*	Impact on nursing hours, administrative burdens, and training overheads.	Adds administrative burden; steep learning curve for frontline nurses.	Neutral impact on hours; manageable onboarding requirements.	Automates routine documentation; directly reduces nursing burnout.
Wait-Time Mitigation *(Weight: 10%)*	Direct reduction of surgical, diagnostic, or specialist backlogs.	No measurable impact on provincial wait-list backlogs.	Localized backlog reduction; small impact on provincial targets.	System-wide bottleneck relief; directly advances provincial targets.

Pillar 2: Fiscal Accountability & Lifecycle Economics (Weight: 30%)

Sub-Metric	Evaluation Criteria	Score 1-4 (Low Impact)	Score 5-7 (Moderate Impact)	Score 8-10 (High Impact)
Cost-Utility & ICER *(Weight: 10%)*	Financial value quantified via Incremental Cost-Effectiveness Ratios.	ICER exceeds $100,000 per QALY; low fiscal justification.	ICER falls between $50,000 and $100,000 per QALY.	ICER is below $50,000 per QALY; clear high-value investment.
Total Cost of Ownership *(Weight: 10%)*	Capture of full 7-year lifecycle costs including hidden overheads.	Significant unquantified costs (e.g., massive data integration fees).	Clear upfront costs; minor variability in long-term maintenance estimates.	Complete transparent visibility into capital, SaaS, and legacy removal costs.
Downstream Cost Offset *(Weight: 10%)*	Ability to prevent expensive acute care or emergency interventions.	High cost with zero proof of downstream system savings.	Displaces some costs, but savings are delayed past a 3-year horizon.	Immediate cash-releasing or capacity-releasing offsets within 12-18 months.

Pillar 3: Health Equity & Jurisdictional Accessibility (Weight: 25%)

Sub-Metric	Evaluation Criteria	Score 1-4 (Low Impact)	Score 5-7 (Moderate Impact)	Score 8-10 (High Impact)
Geographic Parity *(Weight: 10%)*	Technology deployment capability outside the Central Zone (HRM), leveraging remote monitoring or decentralized care.	Central Zone (HRM) bound; relies entirely on urban infrastructure; no remote or decentralized capability.	Deployable to regional hubs; limited remote/decentralized capacity; requires specialized local site preparation.	**Optimized for rural deployment; fully functional across Northern, Western, and Eastern zones; robust remote monitoring.**
Cultural & Structural Inclusivity *(Weight: 10%)*	Explicit design features or implementation plans that actively mitigate health outcome disparities.	Generic design; no targeted strategies for equity-deserving groups; high structural barriers to access.	Standard translation or basic outreach; passive compliance without structural or workflow adaptations.	**Active mitigation of disparities; explicit features co-designed for/with Mi'kmaw, African Nova Scotian, and isolated rural communities.**
Infrastructure Interoperability *(Weight: 5%)*	System compatibility with legacy networks, OPOR architecture, and regional broadband limits.	Standalone silo; incompatible with provincial networks; requires high bandwidth unavailable in rural Nova Scotia.	Basic HL7/FHIR export; manual workarounds needed for OPOR; high bandwidth/cellular network strain.	**Native integration with legacy networks and OPOR architecture; operates efficiently within regional broadband limits.**

Pillar 4: Environmental and System Sustainability (Weight: 15%)

Sub-Metric	Evaluation Criteria	Score 1-4 (Low Impact)	Score 5-7 (Moderate Impact)	Score 8-10 (High Impact)
Environmental Footprint *(Weight: 7.5%)*	Full lifecycle carbon footprint evaluation, hardware efficiency, and medical plastic reduction.	No lifecycle evaluation; heavy reliance on single-use medical plastics; high-energy hardware.	Partial carbon tracking; standard commercial hardware; moderate waste generation throughout the lifecycle.	**Full lifecycle carbon evaluation; energy-efficient hardware; carbon-neutral supply chain; eliminates single-use plastics.**
Contractual & Operational Resilience *(Weight: 7.5%)*	Vendor health, supply chain security for consumables, software update guarantees, and climate/surge adaptability.	Volatile vendor financials; vulnerable supply chains; no extreme weather contingency or update guarantees.	Stable vendor but lacks local stock; standard software SLA; basic surge capacity with some operational risk.	**Strong vendor health; secure consumable supply chains; guaranteed updates; continuous operation during extreme weather/surges.**

Portfolio Scoring Logic & Thresholds

The Final Strategic Score (S) is calculated mathematically using the following formula:

$$S = \sum (Score_i \times Weight_i)$$

Where Score represents the average score (1–10) within a pillar, and Weight represents the allocated pillar percentage.

Strategic Triage Thresholds

- **Score ≥ 85 (Tier 1 - Accelerated Authorization):** Represents flawless alignment with both financial constraints and clinical mandates. Fast-tracked for regional deployment.
- **Score 65–84 (Tier 2 - Conditional Optimization):** Project possesses strong strategic merit but exhibits a deficit in one area (e.g., high clinical value but high cost, or great cost-efficiency but poor rural equity). Requires remediation prior to allocation.
- **Score < 65 (Tier 3 - Deferral / Rejection):** Proposal fails to demonstrate long-term value, system sustainability, or safety parameters. Capital denied.